

MECHANICAL DETERMINANTS OF FASTER CHANGE OF DIRECTION AND AGILITY PERFORMANCE IN FEMALE BASKETBALL ATHLETES

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ABSTRACT

Spiteri, T, Newton, RU, Binetti, M, Hart, NH, Sheppard, JM, and Nimphius, S. Mechanical determinants of faster change of direction and agility performance in female basketball athletes. *J Strength Cond Res* 29(8): 2205–2214, 2015—Change of direction (COD) and agility require the integration of multiple components to produce a faster performance. However, the mechanisms contributing to a faster performance without the confounding factor of athlete expertise or gender is currently unknown. Therefore, the purpose of this study was to assess body composition, strength, and kinetic profile required for a faster COD and agility performance across multiple directional changes. Six faster and 6 slower ($n = 12$) elite female basketball athletes completed a maximal dynamic back squat; eccentric and concentric only back squat; isometric mid thigh pull; whole-body scan to determine lean, fat, and total mass; 505 COD test; T-test; and a multidirectional agility test over in-ground force plates to obtain relevant kinetic measures. Group (faster and slower) by test (2×3) multivariate analyses of variance with follow-up analyses of variance were conducted to examine differences between faster and slower groups and each COD and agility test ($p \leq 0.05$). Faster athletes during the 505 COD test produced significantly greater vertical force ($p = 0.002$) and eccentric and isometric strength capacity ($p = 0.001$). Faster agility and T-test athletes demonstrated significantly shorter contact times ($p = 0.001$), greater propulsive impulse ($p = 0.02$), isometric strength, and relative lean mass compared with slower athletes. Differences between faster athletes across each test seem to be attributed to the mechanical demands of the directional change, increasing force and impulse application as the degree of directional change increased. These findings indicate that different mechanical

properties are required to produce a faster COD and agility performances, and the importance of a greater strength capacity to enable greater mechanical adjustment through force production and body control, during different directional changes.

KEY WORDS force, strength, cutting, decision-making, body composition

INTRODUCTION

Change of direction (COD) and agility maneuvers are multidimensional skills requiring athletes to control individual components (body position, muscle activation, force production, cognitive interpretation) and manipulate the degrees of freedom of the movement to enable constant adaptation within reactive unpredictable environments (27,30,35). This is a particularly important physical quality in basketball, as aggressive directional changes occur throughout a game when athletes compete for positional advantage. Examination of performance times during COD and agility protocols between elite and novice athletes (7,25), and genders (20,29) suggest that elite athletes and males produce a faster COD performance than novice athletes and females, respectively. This difference in performance becomes further evident when combining perception and action during an agility movement, as the task becomes more sport specific (3,20,32). Research has primarily attributed these differences in performance to the type of information-processing strategies used (30,32), lower-body strength components (29,31,36,37), and the subsequent application of force and impulse (31,32), resulting in a more effective and rapid motor response. Although some insight can be drawn from these conclusions, the precise mechanisms contributing to a faster COD or agility performance without the confounding factor of athlete expertise or gender is not yet fully understood.

Differences in ground reaction force (GRF) application have been observed between genders during COD and agility movements, with male athletes applying more force and impulse throughout the movement (29,31,32), and this subsequently produces a faster performance. However, the

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common gender differences observed, specifically those evident in lower-body anthropometrics, body composition, and strength characteristics, can influence the athlete's ability to apply and direct GRFs (1,6,8,32). The results drawn from these collective studies, although focusing on gender differences, provide an indication of determinants of performance that align with comparing stronger and weaker athletes (29). Therefore, the ability to effectively and efficiently use one's strength during COD and agility movements is critical.

When changing direction, athletes must rapidly and systemically coordinate force and impulse application during the braking phase (eccentric), plant phases (isometric), and propulsive phase (concentric) of the movement. Improvement of this performance is often achieved by increasing an athlete's lower-body strength capacity, or more specifically lean muscle mass, which has been shown to improve COD performance (29). Spiteri et al. (32) previously demonstrated that female athletes must possess sufficient eccentric, concentric, dynamic, and isometric strength to enable rapid directional changes, which subsequently increases the amount of force and impulse production throughout the movement. Recent research has demonstrated that greater vertical braking and propulsive force and impulse is required for a faster exit velocity during a 45° COD (29) and offensive and defensive agility movements (31). However, the strength requirements and kinetic profile established for a single 45° directional change may differ for specific basketball COD and agility movements involving different degrees and number of directional changes, as the physical and biomechanical requirements vary between these contexts.

Muscular contribution during COD movements has been found to increase with the number of directional changes and degree of directional change required (36). In basketball, deceleration, reacceleration, and lateral shuffling occur (2,19), which is commonly assessed by the 505 and T-test, providing an indication of an athlete's ability to maneuver around the court. Although these tests replicate the movement patterns commonly observed in basketball, there is a constant integration between perceptual cognitive ability and physical movement throughout the game, often referred to as agility (27). Compared with COD tests, agility tests can involve a different movement patterns that are influenced by the inclusion of an external stimulus (27,35). As such, the force and impulse requirements required to produce a faster performance may vary between the two contexts, depending on the predominant movement mechanics and strength characteristics required to execute the directional change effectively.

Therefore, to determine what biomechanical mechanisms contribute to a faster COD compared with an agility performance, the purpose of this study was to assess the strength components and kinetic profile required for a faster COD and agility performance within female basketball athletes. Magnitudes of GRF, lower-body strength capacity, and body composition profile between faster and slower COD (505 and T-test) and agility performances were

compared. It was hypothesized that athletes who produced a faster COD and agility performance would demonstrate a greater strength capacity and exhibit a higher percentage of lean mass to optimize vertical force and impulse application throughout the movement. Additionally, we also hypothesized that because of the varying mechanical demands of each COD and agility test, the strength requirements, force, and impulse application will differentiate among faster 505, T-test, and agility performances.

METHODS

Experimental Approach to the Problem

This study used a between-subjects design to determine (a) the biomechanical and physical characteristics that differentiate between a faster and slower COD (505 and T-test) and multidirectional agility performance and (b) differences in the biomechanical characteristics between a faster performance in each COD and agility test.

Subjects

Twelve ($n = 12$) female basketball athletes (age: 24.25 ± 2.55 years; height: 177.69 ± 7.25 cm; body mass: 75.56 ± 14.55 kg) playing for a professional basketball team within the Women's National Basketball League (WNBL) were recruited for this study. All athletes were recruited from the same WNBL team consisting of 3 guards, 6 forwards, and 3 centers. To be included for participation within the study, subjects were required to have played basketball for a minimum of 5 years and partake in a minimum of 1 competitive game(s) and 2 structured skills training sessions each week. Data collection occurred after preseason training to ensure adequate fitness and minimal fatigue (as a result from in-season competitive games) for all subjects before the commencement of testing. All subjects were required to be injury free at the time of testing and report no history of major lower limb injuries, such as anterior-cruciate ligament injuries. Ethics approval was obtained from the university's Human Research Ethics Committee before testing, and all testing procedures were explained to the subjects before obtaining their informed consent to participate.

Procedures

Participants were required to attend 3 testing sessions; the first consisted of a 1 repetition maximum (RM) back squat assessment; the second consisted of the first consisted of a dual-energy x-ray absorptiometry (DEXA) scan, isometric midthigh pull, COD (505 and T-test), and agility assessments; and the third consisted of an eccentric and concentric back squat assessment. All testing sessions were separated by 1 week to ensure any fatigue experienced in the previous testing session did not influence the results. All testing occurred before any scheduled training sessions for that week, with a standardized 10-minute dynamic warm-up performed before any testing. Participants were also not allowed to perform any strenuous activity or lower-body resistance training within 48 hours of their assigned testing session.

Dual-Energy X-Ray Absorptiometry. Whole-body scans were performed using DEXA (DXA; Hologic Discovery A, Waltham, MA, USA) to quantify full-body lean, fat, and total mass distribution. Subjects were required to wear minimal clothing with light material, containing no metal (zips, wires, or buttons), and were instructed to remove any metal objects from their pockets before lying in a supine position on the scanning bed with both arms pronated to their side. To ensure consistent and reproducible positioning, subjects were assisted to position their head straight in line with the torso and pelvis, internally rotate and fixate their legs and feet at 45°, and position their arms next to their body within the scanning zone (33). Full-body scan images were subsequently analyzed using manufacturer software (version 12.4; QDR for Windows, Hologic, Waltham, MA, USA) that separated the body into axial and appendicular regions in accordance with the whole-body model (33).

Maximal Dynamic Strength Assessment. Maximal dynamic strength was assessed using a back squat, lowering to a knee flexion angle of 90° (34). Subjects were instructed to position their feet shoulder width apart, toes slightly turned outwards, and position an Olympic bar behind their neck across the trapezius muscle. Knee angle was measured by a manual, handheld goniometer, and an elastic band was placed around the back of the squat rack as an external guide for the required squat depth. Initially, a warm-up was performed consisting of 5–10 repetitions at 40–60% of the subjects' estimated maximum. After a 3-minute rest period, subjects then performed 3–5 repetitions at 60–80% of their perceived maximum. After another 3-minute rest period, the load was subsequently increased with subjects performing 1 repetition at the new load (32,34). Load was continually increased until subjects could not perform the repetition with correct technique (i.e., lowering to the required knee angle) or failed with the weight. A maximum of 5 attempts at any given load was permitted. The maximum load lifted is presented as a value relative to body mass.

Concentric and Eccentric Strength Assessment. Body position was similar to that previously explained for the dynamic back squat. The concentric protocol required subjects to begin seated on a box (box squat), with a knee angle of 90°, extending at the hip and knee to a standing position (28). The eccentric protocol required subjects to flex at the hip and knee from a standing position lowering to a knee angle of 90° (34). Knee angle was monitored as previously described during the maximal dynamic strength protocol. Subjects first commenced the concentric protocol whereby load was added in a linear progression at 60 and 80% of 1RM, performing 2–3 repetitions at each load. Once 80% of the subjects' 1RM was reached, weight was subsequently increased with subjects performing 1 repetition, before increasing to a new load. Weight was continually increased until subjects could not perform the repetition with good

technique or could not lift the required load (28). A rest period of 3 minutes between each set of repetitions was provided. After a 10-minute rest period, subjects then commenced the eccentric protocol that followed the same loading procedures described for the concentric protocol. Both concentric and eccentric movements were performed to a 3-second cadence (12) as set by an external metronome recording. Failure to maintain the 3-second cadence during movements resulted in an unsuccessful completion of that load (12). The maximum load lifted both concentrically and eccentrically for each athlete is presented as a value relative to body mass.

Isometric Strength. Subjects performed a maximal isometric midhigh pull on a portable force plate, sampling at 600 Hz (BP12001200; AMTI, Watertown, NY, USA), with both the knee and hip flexion angles set at 140° (11). Subjects were instructed to drive their feet (positioned shoulder width apart) into the force plate as hard and as fast as they could on commencement of the trial. Subjects were required to perform a total of 3 trials, each trial lasting 5 seconds in duration (22) and separated by a 2-minute recovery period. The force trace for each trial was collected using Bioware software (Version 5x, Type 2812A; Kistler, Switzerland) and exported to Mat-Lab (Version R2010a; Mathworks, Natick, MA, USA) where the average peak force of the 3 trials identified presented as a value relative to body mass ($N \cdot kg^{-1}$).

505 Protocol. The 505 COD test was used within this study as the high velocity 180° directional change executed within the test, replicates a similar movement pattern observed when athletes perform a backdoor cut in basketball. Subjects began standing behind a set of timing gates (Speedlight Timing System, Swift Performance Equipment), before sprinting 10 m through a second set of timing gates, then sprinting a further 5 m (7,32), before contacting their foot on a 600 × 900 mm triaxial force plate (Type 9290AD; Kistler, USA), turning 180°, and completing the test by sprinting 5 m back through the timing gates. Subjects completed 3 trials, planting and changing direction with their preferred leg only. Limb dominance or “preferred limb” was defined as the limb that subjects used as their preferred takeoff foot when performing a lay-up. Approach speed (s) across the first 10 m, and 505 COD time (s), was averaged across the 3 trials for each subject.

T-Test Protocol. The T-test was used within this study to assess how fast athletes can side-shuffle, backpedaling, and forward run, all of which are common movements executed throughout a basketball game to evade or pursue opponents. Subjects began standing behind a set of timing gates (Speedlight Timing System, Swift Performance Equipment), before sprinting forward for 10 m, touching a cone and side shuffling 90° to their left. Subjects then touched a cone before side shuffled a further 10 m to the right, touching

a cone, and side shuffling 5 m back to the left, touching a cone and backpedaling 10 m through timing gates to complete the test (32). Subjects completed 3 trials in total; initiating the first lateral movement on a 600 × 900 mm triaxial force plate (Type 9290AD; Kistler) with their preferred leg. The average time (s) across the 3 trials were determined for each subject.

Multidirectional Agility Test. The multidirectional agility test was developed for this study to assess decision-making time and COD ability when responding to an opponent during 2 closely executed agility movements. Traditionally, agility tests only incorporate 1 directional change in response to a stimulus; however, during sport, multiple directional changes after often required to evade and pursue opponents. As a result, decision-making time and biomechanical strategy may change when multiple directional changes are required. Subjects began behind a set of infrared timing gates (Fitness Technology, Adelaide, Australia) on a marked line 19 m opposite a projection screen and were instructed to run in a straight line toward the projected image (Figure 1). Once the subject reached the timing gates positioned 7.5 m from the starting position, the first visual stimulus (video clip) was programed to automatically start. Subjects then responded to the video by changing direction on a 600 × 900 mm triaxial force plate (Type 9290AD; Kistler), $45 \pm 5^\circ$ to the left or right moving in the same direction as the stimulus (i.e., from a defenders perspective). Once changing direction, subjects then ran a further 3.5 m, triggering timing gates positioned 1.5 m after changing direction, to trigger the second visual stimulus (video clip). Subjects then performed a second directional change, $45 \pm 5^\circ$ either to the left or right in the same direction as the stimulus through timing gates to complete the test. For both directional changes, athletes responded to one of these 8 projected movement patterns (32):

- Change direction by 45° to the left.
- Change direction by 45° to the right.
- Change direction by 45° to the left and pass the ball left.
- Change direction by 45° to the right and pass the ball right.

- Change direction by 45° to the left and pass the ball right.
- Change direction by 45° to the right and pass the ball left.
- Fake right, change direction by 45° to the left, and pass the ball left.
- Fake left, change direction by 45° to the right, and pass the ball right.

These prerecorded videos were triggered by an automated program within the Kinematic Measurement System software (version 13.0; Fitness Technology). Four in-ground force plates positioned in a square layout (Figure 1) were used to ensure capture subjects' foot plant for the first directional change was captured. Two high-speed video cameras (Sony HDD Camcorder HDRXR550V; Sony Australia) sampling at 120 Hz were positioned to the left and right of the agility course adjacent to both directional changes to capture the visual stimulus and the athlete changing direction to determine decision-making time. The average approach velocity ($\text{m} \cdot \text{s}^{-1}$), total running time (s), decision time (s) for the first and second directional change, and running time to complete the first and second directional change (s) was determined across 8 preferred leg trials where the first direction change was performed with the subject planting with their preferred leg. Reliability of this protocol was performed prior testing, resulting in high population-specific test-retest reliability (intraclass correlation coefficient [ICC] = 0.81; Coefficient of variation [CV] = 3.3%).

Data Analysis. Decision-making time during the multidirectional agility test was identified by counting the recorded frames in Silicon Coach (version 6.1.5.1; Silicon-coach Ltd.) as the time between the occlusion of the video stimulus to the first definitive foot plant of the athlete to change direction in response. Raw vertical GRF data were exported to MATLAB programming software (R2010a; The Mathworks Inc., Chatswood, NSW, Australia) to examine specific variables for the preferred limb of the first COD step for each trial during the 505 COD test, T-test, and multidirectional agility test. Variables of interest include relative peak braking and relative propulsive force ($\text{N} \cdot \text{kg}^{-1}$), contact time (s), time spent during the braking and propulsive phase (s), and relative braking and propulsive impulse ($\text{m} \cdot \text{s}^{-1}$). All force and impulse variables were analyzed over the stance phase and calculated relative to body mass and therefore presented as bodyweights, with heel strike defined as the instance the vertical GRF data exceed 10 N, and toe off defined as the instance the

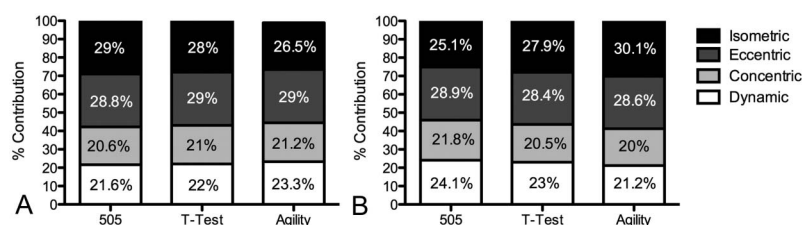


Figure 1. Percent contribution of strength characteristics for (A) faster and (B) slower athletes total strength capacity.

vertical GRF data were below 10 N (20). Braking impulse was calculated from heel strike to the minimum of the midsupport phase, and propulsive impulse was calculated from minimum of midsupport phase to toe off (29).

Statistical Analyses

All results are represented as mean $\pm$ *SD*. An independent T-test was performed to determine differences in performance times (approach speed, total time, and decision-making time) between faster and slower groups for each COD and agility test. A 2×3 multivariate analysis of variance was conducted to examine differences between faster and slower groups and each COD and agility test across all variables. Follow-up 1-way analysis of variance was conducted on each dependent variable to determine precisely where significant differences occurred, with sequential Bonferonni corrections made to reduce type 1 errors (13). A significance level of $p \leq 0.05$ was used throughout all statistical analyses unless otherwise stated. Effect sizes (ESs) were calculated for group comparisons by dividing the difference between groups by the pooled *SD* (5). The magnitude of ES calculations were interpreted following Hopkins (14) guidelines, with trivial = ≤ 0 ; small = 0–0.2; moderate = 0.2–0.6; large = 0.6–1.2; very large = 1.2–2.0; nearly perfect = 2.0–4.0; perfect = ≥ 0.4 . The percentage contribution of each strength assessment was determined for faster and slower groups for each COD and agility test by dividing the average strength assessment score (i.e., average isometric strength) by the total strength score (i.e., the sum of the average maximal dynamic strength, isometric, concentric, eccentric, and power scores for that particular group). All statistical computations were performed using a statistical analysis program (version 17.0; SPSS, Chicago, IL, USA).

RESULTS

Subject characteristics (mean $\pm$ *SD*) for faster and slower groups across both COD and agility test are shown in Table 1. Subjects were separated into faster and slower groups for each COD and agility test based on their total running time achieved during each test. Subjects above the 50th percentile were assigned to the faster group and those below the 50th percentile were assigned to the slower group, similar to previous research (18,29). There were no significant differences in height and age between faster or slower groups ($p = 0.38$ – 0.60 , $ES = 0.7$ – 1.09) or between each COD and agility test ($p = 0.99$ – 1.00 , $ES = 0.58$ – 0.96). Athletes who performed faster during the agility test had significantly lower-body mass ($p = 0.03$; $ES = 2.05$) and significantly greater relative lean mass ($p = 0.04$; $ES = 1.07$) compared with slower athletes. Although percent body fat was generally lower in faster athletes when compared with slower athletes, this difference was only significant between groups for the T-test ($p = 0.03$; $ES = 1.39$). Examination of strength characteristics between faster and slower athletes indicate significantly greater eccentric ($p = 0.01$; $ES = 1.42$) and isometric ($p = 0.02$; $ES = 1.67$) strength for faster athletes

TABLE 1. Subject characteristics (mean $\pm$ *SD*) for faster and slower groups for each COD and agility test.*

	505 COD test		T-test		Agility test	
	Faster (n = 6)	Slower (n = 6)	Faster (n = 6)	Slower (n = 6)	Faster (n = 6)	Slower (n = 6)
Age (y)	24.43 $\pm$ 2.87	22.5 $\pm$ 1.91	24.5 $\pm$ 1.87	22.8 $\pm$ 3.35	24.33 $\pm$ 2.33	23.75 $\pm$ 2.69
Height (cm)	177.38 $\pm$ 7.97	172.28 $\pm$ 6.33	178.83 $\pm$ 8.57	171.56 $\pm$ 3.74	173.8 $\pm$ 6.39	179.05 $\pm$ 9.69
Body weight (kg)	77.32 $\pm$ 5.51	68.3 $\pm$ 4.24	77.75 $\pm$ 4.01	69.6 $\pm$ 6.31	68.5 $\pm$ 5.76†	79.7 $\pm$ 5.12
% Body fat	25.28 $\pm$ 6.13	30.87 $\pm$ 8.22	22.27 $\pm$ 6.80†	31.7 $\pm$ 6.76	23.08 $\pm$ 8.58	29.17 $\pm$ 6.96
Relative lean mass (g·BW ⁻¹)‡	126.01 $\pm$ 7.29	120.53 $\pm$ 13.81	126.61 $\pm$ 12.25	112.03 $\pm$ 6.81	128.41 $\pm$ 12.17†	115.47 $\pm$ 12.04
Dynamic strength (kg·BW ⁻¹)	1.09 $\pm$ 0.32	0.95 $\pm$ 0.17	1.12 $\pm$ 0.34	0.94 $\pm$ 0.14	1.13 $\pm$ 0.29	0.97 $\pm$ 0.26
Concentric strength (kg·BW ⁻¹)	1.03 $\pm$ 0.32	0.86 $\pm$ 0.18	1.07 $\pm$ 0.33	0.84 $\pm$ 0.16	1.03 $\pm$ 0.31	0.92 $\pm$ 0.28
Eccentric strength (kg·BW ⁻¹)	1.44 $\pm$ 0.20†	1.14 $\pm$ 0.22	1.48 $\pm$ 0.42	1.16 $\pm$ 0.20	1.41 $\pm$ 0.39	1.31 $\pm$ 0.37
Isometric strength (N·BW ⁻¹)	1.45 $\pm$ 0.37†	0.99 $\pm$ 0.13	1.41 $\pm$ 0.31†	1.14 $\pm$ 0.26	1.29 $\pm$ 0.39	1.38 $\pm$ 0.39

*BW = bodyweight; COD = change of direction.

†Significant difference ($p \leq 0.05$) in mean values between faster and slower groups.

‡Average relative lean mass in both left and right legs.

TABLE 2. Comparison of performance times (mean $\pm$ SD) between faster and slower groups for each respective COD and agility test.*

	Faster	Slower
505 COD test		
Approach speed (m.s ⁻¹)	2.05 $\pm$ 0.16	2.26 $\pm$ 0.19
COD time (m.s ⁻¹)†	2.43 $\pm$ 0.18	3.03 $\pm$ 0.30
T-test COD test		
Total time (m.s ⁻¹)†	10.36 $\pm$ 0.81	12.75 $\pm$ 0.45
Agility test		
Approach speed (m.s ⁻¹)	1.85 $\pm$ 0.20	1.89 $\pm$ 0.23
First directional change time (m.s ⁻¹)†	1.72 $\pm$ 0.50	2.36 $\pm$ 0.12
Second directional change time (m.s ⁻¹)	0.83 $\pm$ 0.22	1.02 $\pm$ 0.11
Total time (m.s ⁻¹)†	4.47 $\pm$ 0.62	5.34 $\pm$ 0.25
First COD decision-making time (m.s ⁻¹)†	0.15 $\pm$ 0.03	0.20 $\pm$ 0.12
Second COD decision-making time (m.s ⁻¹)	0.17 $\pm$ 0.11	0.21 $\pm$ 0.09

*COD = change of direction.
†Significant difference ($p \leq 0.05$) in mean values between faster and slower groups.

each strength measure for the selected COD and agility test are very similar for both faster and slower athletes (Figure 1).

The comparison of performance times indicated that faster athletes demonstrated a significantly faster COD time ($p = 0.01$; ES = 2.42) during the 505, and total time ($p = 0.01$, ES = 3.64) during the T-test when compared with slower athletes (Table 2). For the agility test, faster athletes demonstrated a significantly faster first COD time ($p = 0.04$; ES = 0.43), total time ($p = 0.04$; ES = 1.84), and decision-making time ($p = 0.03$; ES = 0.57) for the first directional change, whereas

during the 505 COD test, whereas significantly greater isometric strength ($p = 0.02$; ES = 0.94) was characteristic of faster athletes when compared with slower athletes during the T-test (Table 1). Although faster athletes demonstrated an overall larger strength capacity, the percent contribution of

no significant difference was observed between faster and slower athletes for approach speed ($p = 0.62$; ES = 0.18), second COD time ($p = 0.23$; ES = 1.09), or decision-making time ($p = 0.07$; ES = 1.03) for the second directional change.

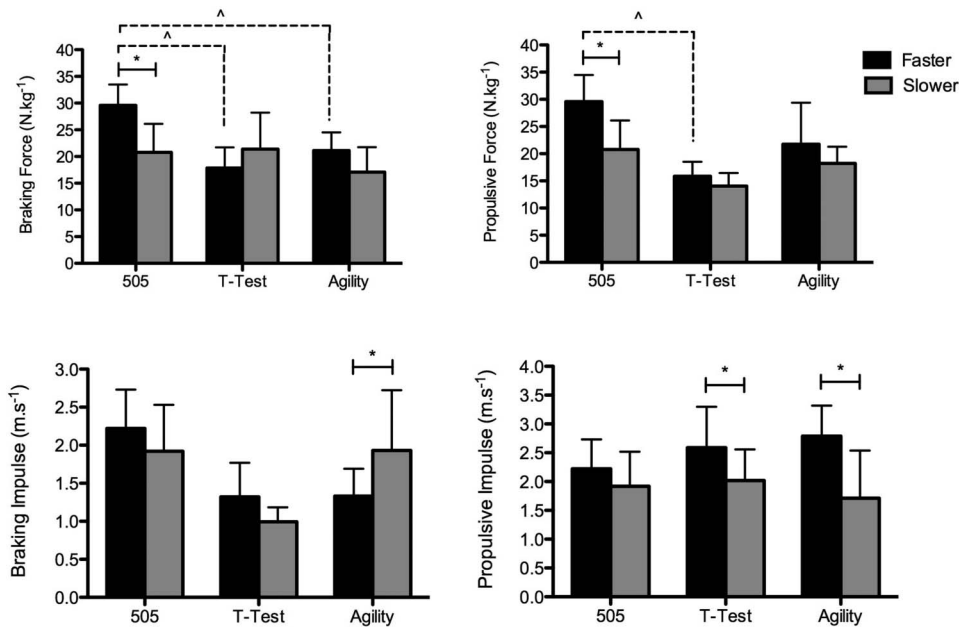


Figure 2. Comparison of vertical braking force, propulsive force, braking impulse, and propulsive impulse between faster and slower athletes across each COD and agility test. COD = change of direction.

*Significant difference ($p \leq 0.05$) in mean values between faster and slower groups.
^Significant difference ($p \leq 0.05$) in mean values between faster performers during the COD and agility tests.

TABLE 3. Timing variables between COD and agility tests.*

	505 COD test		T-test		Agility test	
	Faster (n = 6)	Slower (n = 6)	Faster (n = 6)	Slower (n = 6)	Faster (n = 6)	Slower (n = 6)
Contact time (s)†	0.42 ± 0.03‡	0.47 ± 0.04	0.32 ± 0.03‡†	0.35 ± 0.03	0.42 ± 0.04‡	0.51 ± 0.05
Time spent braking (s)†	0.17 ± 0.02‡	0.19 ± 0.03	0.09 ± 0.03	0.13 ± 0.05	0.08 ± 0.02‡	0.17 ± 0.25
Time spent propulsive (s)†	0.26 ± 0.03‡	0.27 ± 0.04	0.19 ± 0.02	0.25 ± 0.05	0.16 ± 0.06‡	0.17 ± 0.02

*COD = change of direction.

†Significant difference ($p \leq 0.05$) in mean values between faster and slower groups.

‡Significant difference ($p \leq 0.05$) in mean values between faster performers during the COD and agility tests.

Comparison of vertical braking and propulsive force and impulse between faster and slower athletes are shown in Figure 2. Faster athletes during the 505 COD test displayed significantly greater vertical braking force compared with slower athletes during the 505 ($p = 0.02$; ES = 1.88), and compared with faster athletes during the T-test ($p = 0.01$; ES = 3.02) and agility test ($p = 0.02$, ES = 2.31). Vertical propulsive force was significantly different between faster and slower athletes during the 505 COD test ($p = 0.02$; ES = 1.72), and between faster athletes during the T-test ($p = 0.001$; ES = 3.50) and agility test ($p = 0.03$; ES = 2.85). Differences in vertical braking impulse was only observed during the agility test, with slower athletes producing significantly greater braking impulse compared with faster athletes ($p = 0.04$; ES = 0.53). Vertical propulsive impulse was significantly greater for faster athletes during both the T-test ($p = 0.03$; ES = 0.91) and agility test ($p = 0.02$; ES = 1.55) when compared with slower athletes.

Examination of ground contact times during the plant phase revealed faster athletes across all COD and agility tests produced shorter contact times when compared with slower athletes (Table 3). Furthermore, faster athletes during the T-test demonstrated significantly shorter contact times when compared with slower athletes in both the 505 COD test and agility test ($p = 0.001$; ES = 3.33). Faster athletes during the agility test demonstrated a significantly shorter braking time when compared with slower athletes ($p = 0.001$; ES = 0.51). Significantly, longer braking time ($p = 0.001$; ES = 3.3) and propulsive time ($p = 0.001$; ES = 2.43) were observed for faster athletes during the 505 COD test when compared with both the agility and T-test.

DISCUSSION

This study is the first to examine the differences in vertical GRF and impulse variables, strength capacity, and body composition between faster and slower female athletes during basketball-specific COD and agility movements and compare differences between each COD and agility test to determine the characteristics required for a faster perfor-

mance. The findings from this study demonstrated that differences exist between force-dependent variables (505 COD test), kinetic timing variables (T-test and agility test), strength characteristics (505 COD test and T-test), and body composition measures (T-test and agility) between faster and slower athletes. When examining differences between faster athletes in each COD and agility test, the predominant difference seems to be attributed to the mechanical demands of the required direction change, with faster athletes in the 505 COD test producing greater force and longer contact times compared with both the T-test and agility test. These findings demonstrate that different mechanical properties are required to produce a faster COD and agility performance, which should be developed to improve performance for the multiple directional changes required in basketball.

It was recently established that greater force production when changing direction is a combination of superior movement mechanics and strength capacity, resulting in a faster COD performance (29,32). This finding is supported by this study, with faster athletes in the 505 COD test producing significantly greater braking and propulsive force compared with slower athletes (Figure 2). Increasing force application during the braking phase of COD movements has been shown to increase exit velocity during COD movements (9,10,29) due to an increased storage and utilization of elastic energy as the muscle lengthens under an eccentric load (15,29). As no significant difference was observed in approach velocity between faster and slower athletes, the greater braking force application observed in faster athletes is a direct result of an increased eccentric strength capacity to accept and apply force during this phase (9,16,32). Greater eccentric strength enabled faster athletes to complete the direction change within a significantly shorter braking time enabling a faster transition into the propulsive phase of the movement, and increasing propulsive force application (9,29). Additionally, faster athletes demonstrated significantly greater isometric strength, which is essential to maintain a lower-body position during the braking and propulsive phase of the movement (24), which optimizes triple

extension of the lower body. This allows athletes to control the displacement of their body to successfully transfer force in the new direction to produce a faster COD movement.

The high velocity 180° directional change as observed in the 505, replicates a backdoor cut in basketball and simulates the requirements of athletes to position themselves between their opponents. However, the maneuverability as observed in the T-test requires athletes to reposition and shift their momentum in multiple directions, increasing the muscular demands of the lower body to constantly decelerate and reaccelerate (36). Although faster athletes displayed an increased strength capacity compared with slower athletes, only isometric strength was significantly greater for faster performers during the T-test. Similar to the 505 COD test, greater isometric strength enables athletes to maintain and lower into a defensive ready position (24,32), and improving mechanical functioning of the lower limbs by optimizing the muscles length-tension relationship to increase force output and acceleration ability. This finding was supported as shorter braking, propulsive and contact times were observed for faster athletes. Additionally, faster athletes displayed a significantly lower-body fat percentage, assisting them to perform the COD movement faster as a result of less nonfunctional mass (23). Although no difference was observed in braking force and impulse application, faster athletes produced significantly greater propulsive impulse compared with slower athletes. Although we have previously discussed the neuromuscular advantages of greater braking force and impulse (9,15,29), the T-test involves a 90° directional change, which would likely require less braking force to shift their momentum in a lateral direction. Therefore, rapid deceleration is not a distinguishing factor between faster and slower performers in the T-test. Instead, it is the ability to maintain a lower-body position, which can allow the athlete to rapidly extend at the hip, which increases propulsive ability.

Basketball involves multiple offensive and defensive scenarios, requiring athletes to make correct on-court decisions and possess the required physical and technical attributes to adjust their body position and react to stimuli in the surrounding environment. Findings from this study reveal integration between perceptual cognitive ability and the biomechanical components is required to produce a faster agility performance. However, while no difference was observed in braking force application, slower athletes produced significantly greater braking impulse through a longer ground contact time during the agility plant phase. Braking impulse has been found to improve acceleration ability through greater force application (9,15,29). However, increasing braking impulse through a longer contact time in game scenarios is not advantageous when the purpose is to evade or pursue opponents in a time-restricted manner. Shorter braking contact time, in conjunction with a quicker first COD decision-making time, enabled faster athletes to produce a quicker first COD time. This indicated that once athletes are able to identify relevant body kinematic cues

(20,25,27,31,32), faster acceleration could occur through greater propulsive impulse application. Furthermore, while no difference was observed in strength between faster or slower agility groups, faster athletes displayed significantly greater lean muscle mass and lower total body mass. Numerous studies have observed weak correlations between strength measures and agility performance (17,26,32,37). However, carrying more nonfunctional mass may result in a disadvantage resulting in a slower athlete, particularly where there is a requirement for rapid changes in momentum (23). The additional nonfunctional mass would negatively affect the ability to change direction, particularly in a time sensitive movement like agility.

Although faster athletes demonstrated a significantly faster first decision-making time during the multidirectional agility test, no difference was observed in second decision-making time. This resulted in an equivalent second COD time between groups (Table 2). As ground kinetics were not obtained for the second COD movement, we cannot definitively conclude if the slower second COD time was due to a delay in cognitive or physical processes. Previous research investigating reaction time during a dual-stimulus has observed a delay in response to a second stimulus, as the first stimulus is still being processed (4). This phenomenon, termed the psychological refractory period, interferes with the movement programming stage of the information-processing model delaying subsequent movement output (21). The agility test in this study involved 2 directional changes; the second occurring in close proximity to the first directional change, to replicate the reactive environment of basketball. As cognitive delay has been shown to affect movement output (4,7,27,32), the amount of preactivation and subsequent force output for the second directional change would be compromised. Therefore, while faster athletes produced a faster agility performance as a result of a faster first decision-making time and first COD time, it seems perceptual cognitive factors greatly influence subsequent COD performance when athletes are required to respond to multiple stimuli.

Basketball athletes perform multiple directional changes throughout a game, requiring different biomechanical, physical, and perceptual cognitive abilities to produce a fast COD movement. Regardless of how fast an athlete can perform the 505 COD test, time spent during the braking and propulsive phase of the movement is significantly longer compared with faster performers in the T-test and agility test (Table 3). This time difference is the direct result of mechanical differences between these COD movements. When athletes perform a 180° backdoor cut, similar to the 505, a greater eccentric load and braking capacity is required to stop suddenly from a high velocity approach to transfer this momentum in the new direction. This increases braking force application due to the angle of directional change, requiring athletes to move through a greater range of motion and reorientate their body position to optimize full extension

of the hip, knee, and ankle to increase propulsive force application (3,20,31,32). Although both the T-test and agility test require a greater propulsive force application to produce a faster COD movement, athletes maintain a more upright position during these directional changes (24), decreasing the range of motion required to reaccelerate, resulting in a reduced propulsive force application.

When examining the contribution of strength characteristics, no difference was observed between COD and agility tests. Although studies have shown correlations between eccentric, concentric, and isometric strength to COD performance (1,16,32,36), the percent contribution of each strength measure to both COD and agility performance is extremely similar (Figure 1). This demonstrates that all strength qualities are used during COD and agility movements. More importantly, the study results revealed the importance of developing a greater strength capacity across all strength qualities within female athletes to produce a faster COD movement, instead of redistribution of strength quality development.

Change of direction and agility movements are a multidimensional skill involving numerous variables to produce a faster performance (27). Factors such as technique, muscle activation, and visual search strategies were not investigated and are limitations of this study, as these factors also influence an athlete's ability to produce a faster COD and agility movement. Although these findings are limited to a small ($n = 12$) population of elite female basketball athletes, the results of this study are from a homogenous group with a similar training background, playing experience, and current training schedules, making the findings less variable, and thereby adding a degree of strength to the analysis. In conclusion, within the context of these limitations, this study demonstrated that faster athletes in each respective COD and agility test had a different kinetic, strength, and physical profile compared with slower athletes, which resulted in a superior performance. Furthermore, when comparing faster performers across each test, differences in kinetic variables seem to be attributed to the mechanical demands (i.e., the angle of directional change) of the movement.

PRACTICAL APPLICATIONS

These findings demonstrate the importance of applying greater force and impulse, while controlling body position when changing direction to achieve a faster performance. Although significant differences were observed in eccentric strength (505 COD test) and isometric strength (505 COD test and T-test) between faster and slower athletes, the percentage contribution of each strength measure for faster and slower groups across each COD and agility test were extremely similar. However, when examining differences between directional changes, such as a 90° shuffle, a 180° backdoor cut and a 45° directional change, the emphasis and component of strength required to decelerate, and maintain appropriate body positioning throughout the movement differs. Together, these findings highlight the importance of

developing all components of strength and increasing lean muscle mass in female athletes. This will enable athletes to apply a greater amount of force and impulse through the movement as a result of an increased strength capacity, without increasing contact time, which is important for a faster COD and agility performance. Additionally, developing athlete's perceptual cognitive ability and reaction to multiple stimuli through exercises, such as small-sided games and shadow drills, will assist in developing effective cue recognition to improve movement sequencing multiple directional changes. Improvement in athletes' ability to identify relevant cues earlier in the stimulus-presentation phase will allow them to decelerate sooner by applying greater force and impulse during the braking phase of the movement, resulting in a shorter contact time and faster agility movement.

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REFERENCES

1. Baker, DG and Newton, R. Comparison of lower body strength, power, acceleration, speed, agility, and sprint momentum to describe and compare playing rank among professional rugby league players. *J Strength Cond Res* 22: 153–158, 2008.
2. Balciunas, M, Stonkus, S, Abrantes, C, and Sampaio, J. Long term effects of different training modalities on power, speed, skill and anaerobic capacity in young male basketball players. *J Sports Sci Med* 5: 163–170, 2006.
3. Bradshaw, RJ, Young, WB, Russell, A, and Burge, P. Comparison of offensive agility techniques in Australian Rules football. *J Sci Med Sport* 14: 65–69, 2011.
4. Brisslewalter, J, Collardeau, M, and Rene, A. Effects of acute physical exercise characteristics on cognitive performance. *Sports Med* 32: 555–566, 2002.
5. Cohen, J. *Statistical Power Analysis for the Behavioural Sciences* (2nd ed.). Hillsdale, NJ: Lawrence Erlbaum Associates, 1988.
6. Erculj, F, Blas, M, and Barcia, M. Physical demands on young elite European female basketball players with special reference to speed, agility, explosive strength and take-off power. *J Strength Cond Res* 24: 2970–2978, 2010.
7. Gabbett, TJ, Kelly, JN, and Sheppard, JM. Speed, change of direction speed, and reactive agility of rugby league players. *J Strength Cond Res* 22: 174–181, 2008.
8. Garstecki, MA, Latin, RW, and Cuppett, MM. Comparison of selected physical fitness and performance variables between NCAA Division I and II football players. *J Strength Cond Res* 18: 292–297, 2004.
9. Glaister, BC, Orenduff, MS, Schoen, JA, and Bernatz, GC. Ground reaction forces and impulses during a transient maneuver. *J Biomech* 41: 3090–3093, 2008.
10. Green, BS, Blake, C, and Caulfield, BM. A comparison of cutting technique performance in rugby union players. *J Strength Cond Res* 25: 2668–2680, 2011.
11. Haff, GG, Jackson, JR, Kawamori, N, Carlock, JM, Hartman, MJ, Kilgore, JL, Morris, RT, Ramsey, MW, Sands, WA, and Stone, MH. Force-time curve characteristics and hormonal alterations during an eleven-week training period in elite women weightlifters. *J Strength Cond Res* 22: 433–446, 2008.

12. Hollander, DB, Kraemer, RR, Kilpatrick, MW, Ramadan, ZG, Reeves, GV, Francois, M, Herbert, EP, and Tryniecki, JL. Maximal eccentric and concentric strength discrepancies between young men and women for dynamic resistance exercise. *J Strength Cond Res* 21: 37–40, 2007.
13. Holm, S. A simple sequentially rejective multiple test procedure. *Scand J Statist* 6: 65–70, 1979.
14. Hopkins, WG. A new view of statistics. *Internet Soc Sport Sci* 2002. Available at: <http://www.sportsci.org/resource/stats>. Accessed June 2, 2014.
15. Hunter, JP, Marshall, RN, and McNair, PJ. Relationship between ground reaction force impulse and kinematics of sprint-running acceleration. *J Appl Biomech* 21: 31–43, 2005.
16. Jones, P, Bampouras, T, and Marrin, K. An investigation into the physical determinants of change of direction speed. *J Sports Med Phys Fitness* 49: 97–104, 2009.
17. Markovic, G. Poor relationship between strength and power qualities and agility performance. *J Sports Med Phys Fitness* 47: 276–283, 2007.
18. McBride, JM, Nimphius, S, and Erikson, TM. The acute effects of heavy-load squats and loaded countermovement jumps on sprint performance. *J Strength Cond Res* 19: 893–898, 2005.
19. McInnes, S, Carlson, J, Jones, C, and McKenna, M. The physiological loads imposed on basketball players during competition. *J Sport Sci* 5: 387–397, 1995.
20. McLean, SG, Lipfert, SW, and Van Den Bogert, AJ. Effect of gender and defensive opponent on the biomechanics of sidestep cutting. *Med Sci Sports Exerc* 36: 1008–1016, 2004.
21. McMorris, T. *Acquisition and Performance of Sports Skills*. Chichester, England: Wiley, 2004.
22. Nuzzo, J, McBride, J, Cormie, P, and McCauley, G. Relationship between countermovement jump performance and multijoint isometric and dynamic tests of strength. *J Strength Cond Res* 22: 669–707, 2008.
23. Ostojic, SM, Mazic, S, and Dikic, N. Profiling in basketball: Physical and physiological characteristics of elite players. *J Strength Cond Res* 20: 740–744, 2006.
24. Sasaki, S, Nagano, Y, Kaneko, S, Sakurai, T, and Fukubayashi, T. The relationship between performance and trunk movement during change of direction. *J Sports Sci Med* 10: 112–118, 2011.
25. Serpell, BG, Ford, M, and Young, WB. The development of a new test of agility for rugby league. *J Strength Cond Res* 24: 3270–3277, 2009.
26. Scanlan, A, Humphries, B, Tucker, PS, and Dalbo, V. The influence of physical and cognitive factors on reactive agility performance in men basketball players. *J Sports Sci* 32: 367–374, 2013.
27. Sheppard, JM and Young, WB. Agility literature review: Classifications, training and testing. *J Sports Sci* 24: 919–932, 2006.
28. Sleivert, G and Taingahue, M. The relationship between maximal jump-squat power and sprint acceleration in athletes. *Eur J Appl Physiol* 91: 46–52, 2004.
29. Spiteri, T, Cochrane, JL, Hart, NH, Haff, GG, and Nimphius, S. Effect of strength on plant foot kinetics and kinematics during a change of direction task. *Eur J Sport Sci* 13: 646–652, 2013.
30. Spiteri, T, Cochrane, JL, and Nimphius, S. The evaluation of a new lower body reaction time test. *J Strength Cond Res* 27: 174–180, 2012.
31. Spiteri, T, Hart, NH, and Nimphius, S. Offensive and defensive agility: A sex comparison of lower body kinematics and ground reaction forces. *J Appl Biomech* 30: 514–520, 2014.
32. Spiteri, T, Nimphius, S, Specos, C, Hart, NH, Sheppard, JM, and Newton, RU. The contribution of strength characteristics to change of direction and agility in female basketball athletes. *J Strength Cond Res* 28: 2415–2423, 2014.
33. Stewart, AD and Hannan, WJ. Prediction of fat and fat-free mass in male athletes using dual X-ray absorptiometry as the reference method. *J Sports Sci* 18: 263–274, 2000.
34. Yetter, M and Moir, GL. The acute effects of heavy back and front squats on speed during forty-meter sprint trials. *J Strength Cond Res* 22: 159–165, 2008.
35. Young, WB and Farrow, D. A review of Agility: Practical applications for strength and Conditioning. *Strength Cond J* 28: 24–29, 2006.
36. Young, WB, James, R, and Montgomery, I. Is muscle power related to running speed with changes of direction?. *J Sports Med Phys Fitness* 42: 282–288, 2002.
37. Young, WB, Miller, I, and Talpey, S. physical qualities predict change-of-direction speed but not defensive agility in Australian rules football. *J Strength Cond Res* 2014. Epub ahead of print.